

## FCFS CODE:

```
#include <stdio.h>

struct Process {
    int pid, at, bt;
    int ct, tat, wt, rt;
};

int main() {

    printf("riya gupta, 1WA24CS235");
    int n;
    printf("Enter number of processes: ");
    scanf("%d", &n);

    struct Process p[n];

    for(int i = 0; i < n; i++) {
        p[i].pid = i + 1;
        printf("\nProcess %d\n", p[i].pid);
        printf("Arrival Time: ");
        scanf("%d", &p[i].at);
        printf("Burst Time: ");
        scanf("%d", &p[i].bt);
    }

    int time = 0;
    float avg_tat = 0, avg_wt = 0, avg_rt = 0;

    for(int i = 0; i < n; i++) {

        if(time < p[i].at)
            time = p[i].at;

        p[i].rt = time - p[i].at;
        time += p[i].bt;

        p[i].ct = time;
        p[i].tat = p[i].ct - p[i].at;
        p[i].wt = p[i].tat - p[i].bt;

        avg_tat += p[i].tat;
```

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        avg_wt += p[i].wt;
        avg_rt += p[i].rt;
    }

    printf("\nPID\tAT\tBT\tCT\tTAT\tWT\tRT\n");
    for(int i = 0; i < n; i++) {
        printf("%d\t%d\t%d\t%d\t%d\t%d\t%d\n",
            p[i].pid, p[i].at, p[i].bt,
            p[i].ct, p[i].tat, p[i].wt, p[i].rt);
    }

    printf("\nAverage Turnaround Time = %.2f", avg_tat/n);
    printf("\nAverage Waiting Time = %.2f", avg_wt/n);
    printf("\nAverage Response Time = %.2f", avg_rt/n);

    return 0;
}
```

riya gupta, 1WA24CS235Enter number of processes: 3

Process 1

Arrival Time: 2

Burst Time:

5

Process 2

Arrival Time: 3

Burst Time: 2

Process 3

Arrival Time: 4

Burst Time: 3

| PID | AT | BT | CT | TAT | WT | RT |
|-----|----|----|----|-----|----|----|
| 1   | 2  | 5  | 7  | 5   | 0  | 0  |
| 2   | 3  | 2  | 9  | 6   | 4  | 4  |
| 3   | 4  | 3  | 12 | 8   | 5  | 5  |

Average Turnaround Time = 6.33

Average Waiting Time = 3.00

Average Response Time = 3.00

=== Code Execution Successful ===

## SJF(NON-PRIORITY) CODE:

```
#include <stdio.h>

struct Process {
    int pid, at, bt;
    int ct, tat, wt, rt;
    int done;
};

int main() {
    printf("riya gupta, 1WA24CS235");
    int n;
    printf("Enter number of processes: ");
    scanf("%d", &n);

    struct Process p[n];

    for(int i = 0; i < n; i++) {
        p[i].pid = i + 1;
        printf("\nProcess %d\n", p[i].pid);
        printf("Arrival Time: ");
        scanf("%d", &p[i].at);
        printf("Burst Time: ");
        scanf("%d", &p[i].bt);
        p[i].done = 0;
    }

    int time = 0, completed = 0;
    float avg_tat = 0, avg_wt = 0, avg_rt = 0;

    while(completed < n) {
        int min = -1;

        for(int i = 0; i < n; i++) {
            if(!p[i].done && p[i].at <= time) {
                if(min == -1 || p[i].bt < p[min].bt)
                    min = i;
            }
        }

        if(min == -1) {
            time++;
        } else {
            p[min].done = 1;
            completed++;
            avg_tat += p[min].tat;
            avg_wt += p[min].wt;
            avg_rt += p[min].rt;
            time += p[min].bt;
        }
    }

    printf("Average TAT: %.2f\n", avg_tat/n);
    printf("Average WT: %.2f\n", avg_wt/n);
    printf("Average RT: %.2f\n", avg_rt/n);
}
```

```

    p[min].rt = time - p[min].at;
    time += p[min].bt;

    p[min].ct = time;
    p[min].tat = p[min].ct - p[min].at;
    p[min].wt = p[min].tat - p[min].bt;
    p[min].done = 1;
    completed++;

    avg_tat += p[min].tat;
    avg_wt += p[min].wt;
    avg_rt += p[min].rt;
}
}

printf("\nPID\tAT\tBT\tCT\tTAT\tWT\tRT\n");
for(int i = 0; i < n; i++) {
    printf("%d\t%d\t%d\t%d\t%d\t%d\t%d\n",
        p[i].pid, p[i].at, p[i].bt,
        p[i].ct, p[i].tat, p[i].wt, p[i].rt);
}

printf("\nAverage Turnaround Time = %.2f", avg_tat/n);
printf("\nAverage Waiting Time = %.2f", avg_wt/n);
printf("\nAverage Response Time = %.2f", avg_rt/n);

return 0;
}

```

## Output

[Clear](#)

riya gupta, 1WA24CS235Enter number of processes: 4

Process 1

Arrival Time: 3

Burst Time: 4

Process 2

Arrival Time: 2

Burst Time:

6

Process 3

Arrival Time: 1

Burst Time: 4

Process 4

Arrival Time: 3

Burst Time: 7

| PID | AT | BT | CT | TAT | WT | RT |
|-----|----|----|----|-----|----|----|
| 1   | 3  | 4  | 9  | 6   | 2  | 2  |
| 2   | 2  | 6  | 15 | 13  | 7  | 7  |
| 3   | 1  | 4  | 5  | 4   | 0  | 0  |
| 4   | 3  | 7  | 22 | 19  | 12 | 12 |

Average Turnaround Time = 10.50

Average Waiting Time = 5.25

Average Response Time = 5.25

## SJF(PRIORITY) CODE:

```
#include <stdio.h>

struct Process {
    int pid, at, bt;
    int ct, tat, wt, rt;
    int remaining;
    int started;
};

int main() {
    printf("riya gupta, 1WA24CS235");

    int n;
    printf("Enter number of processes: ");
    scanf("%d", &n);

    struct Process p[n];

    for(int i = 0; i < n; i++) {
        p[i].pid = i + 1;
        printf("\nProcess %d\n", p[i].pid);
        printf("Arrival Time: ");
        scanf("%d", &p[i].at);
        printf("Burst Time: ");
        scanf("%d", &p[i].bt);
        p[i].remaining = p[i].bt;
        p[i].started = 0;
    }

    int time = 0, completed = 0;
    float avg_tat = 0, avg_wt = 0, avg_rt = 0;

    while(completed < n) {
        int min = -1;

        for(int i = 0; i < n; i++) {
            if(p[i].at <= time && p[i].remaining > 0) {
                if(min == -1 || p[i].remaining < p[min].remaining)
                    min = i;
            }
        }
    }
}
```

```

if(min == -1) {
    time++;
} else {

    if(p[min].started == 0) {
        p[min].rt = time - p[min].at;
        p[min].started = 1;
    }

    p[min].remaining--;
    time++;

    if(p[min].remaining == 0) {
        p[min].ct = time;
        p[min].tat = p[min].ct - p[min].at;
        p[min].wt = p[min].tat - p[min].bt;
        completed++;

        avg_tat += p[min].tat;
        avg_wt += p[min].wt;
        avg_rt += p[min].rt;
    }
}

printf("\nPID\tAT\tBT\tCT\tTAT\tWT\tRT\n");
for(int i = 0; i < n; i++) {
    printf("%d\t%d\t%d\t%d\t%d\t%d\t%d\n",
        p[i].pid, p[i].at, p[i].bt,
        p[i].ct, p[i].tat, p[i].wt, p[i].rt);
}

printf("\nAverage Turnaround Time = %.2f", avg_tat/n);
printf("\nAverage Waiting Time = %.2f", avg_wt/n);
printf("\nAverage Response Time = %.2f", avg_rt/n);

return 0;
}

```

## Output

[Clear](#)

riya gupta, 1WA24CS235Enter number of processes: 3

Process 1

Arrival Time: 2

Burst Time: 5

Process 2

Arrival Time: 3

Burst Time: 2

Process 3

Arrival Time: 1

Burst Time:

4

| PID | AT | BT | CT | TAT | WT | RT |
|-----|----|----|----|-----|----|----|
| 1   | 2  | 5  | 12 | 10  | 5  | 5  |
| 2   | 3  | 2  | 5  | 2   | 0  | 0  |
| 3   | 1  | 4  | 7  | 6   | 2  | 0  |

Average Turnaround Time = 6.00

Average Waiting Time = 2.33

Average Response Time = 1.67

=== Code Execution Successful ===